

The Architectural Gap in Current AI: A Derivation from the Laws of Information Conservation, Connection, and Autoinjection

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Abstract

Current artificial intelligence systems exhibit characteristic failure modes: context-bounded memory loss, inability to navigate previously failed reasoning paths, and the absence of genuine structural self-evaluation. This paper argues that these failures are not incidental engineering limitations but consequences of a fundamental architectural gap: current AI systems are built without reference to the foundational laws governing information. Drawing on the three foundational laws of the Special Theory of Everything (SToE) — the Law of Information Conservation, the Law of Connection, and the Law of Autoinjection — we derive the specific architectural requirements that a genuinely information-native AI system must satisfy. We describe the required vessel: a persistent reasoning graph in which no information point is discarded, a topology-aware navigation mechanism distinct from similarity-based retrieval, and a structural self-evaluation loop grounded in autoinjection. We assess the current state of AI development against each requirement, identify the precise gaps, and restate the testable prediction that distinguishes an information-conserving architecture from existing approaches. The three laws function not as philosophical commentary on AI but as a formal design specification for the next generation of AI architecture.

Keywords: Special Theory of Everything, information conservation, autoinjection, AI architecture, reasoning graph, persistent memory, self-evaluation, foundational laws

1. Introduction

Contemporary artificial intelligence has achieved remarkable results in pattern recognition, language generation, code synthesis, and multi-step reasoning within bounded contexts. Yet practitioners and researchers consistently identify a cluster of persistent failure modes that

improvements in scale and training have not eliminated: coherence degrades over long reasoning chains; information introduced early in a session becomes inaccessible as the context window fills; failed reasoning paths are discarded rather than retained; self-evaluation is simulated through prompting rather than grounded in architecture. These failures are typically treated as engineering problems awaiting engineering solutions — larger context windows, better retrieval systems, improved chain-of-thought prompting.

This paper argues that these failures have a common source that is not engineering but foundational. Current AI systems are built without reference to the fundamental laws governing information itself. As a result, their architectures systematically violate three properties that any system working with information must respect: conservation, observer-dependent connection, and autoinjective self-containment. The failures that practitioners observe are the predictable consequences of these violations.

The three foundational laws of the Special Theory of Everything (SToE), developed across Voronin (2021, 2025a, 2025b) and stated explicitly in Voronin (2025c), provide the formal basis for this argument. These laws — the Law of Information Conservation, the Law of Connection, and the Law of Autoinjection — are not philosophical metaphors. They are structural commitments that generate specific, derivable architectural requirements. The purpose of this paper is to make that derivation explicit: to show precisely what an AI architecture must look like if it is to operate in accordance with these laws, to assess where current approaches stand relative to each requirement, and to specify the vessel that is needed.

The argument proceeds as follows. Section 2 restates the three laws as design constraints. Section 3 derives the architectural requirements from each law. Section 4 assesses the current state of AI development against these requirements and identifies the precise gaps. Section 5 describes the required vessel in full. Section 6 restates the testable prediction that distinguishes an information-conserving architecture from existing approaches. Section 7 considers the relation to existing research directions. Section 8 concludes.

2. The Three Laws as Design Constraints

The three foundational laws of SToE were stated in Voronin (2025c). They are reproduced here in the form most relevant to architectural derivation.

2.1 The Law of Information Conservation

Any information point does not vanish from reality. It can only lose connections with other information points.

What appears as loss — forgetting, deletion, error, discarded output — is not annihilation of an information point but severance of its connections to the currently reachable network. The information point persists in the field; it becomes unreachable from the connected region accessible to a given observer. The field contains no mechanism of deletion.

As a design constraint, this law prohibits any architecture that genuinely discards information. Compression, summarization, and context truncation all constitute violations if

they result in information that cannot be recovered through any traversal of the system. The distinction between annihilation and disconnection is architectural, not merely terminological: a conserving system must maintain the information point in retrievable form, even if it is currently outside the active processing region.

2.2 The Law of Connection

Connections between information points are created only by an observer or a creator.

Information points do not spontaneously connect. Connection requires an agent — an observer that perceives or names another information point, or a creator that generates new information points. The set of information points reachable from any given observer is determined by the chain of connections that observer can establish or traverse.

As a design constraint, this law specifies that navigation through the information field requires active agency. A system cannot passively retrieve conserved information points; it must have a mechanism for establishing connections between its current state and previously conserved points. Similarity-based retrieval approximates this but does not satisfy it: similarity search identifies points whose content resembles the query, not points that are structurally adjacent in the reasoning topology. The law requires a navigation mechanism that understands connection structure, not merely content similarity.

2.3 The Law of Autoinjection

Everything is an information point, including the information field itself. There is no exterior to the system.

The information field is closed and self-containing. Every observer, including the system performing reasoning, is an information point within the field. There is no external standpoint from which to evaluate the field. This means that self-evaluation is not an optional add-on but a structural necessity: a system operating within an autoinjective field must be capable of receiving itself as input, evaluating its own outputs, and treating those evaluations as information points subject to the same conservation and connection laws.

As a design constraint, this law requires that self-evaluation be grounded in architecture rather than simulated through prompting. A system that evaluates its outputs by querying itself with a meta-prompt is not autoinjective in the formal sense: the evaluation is not structurally connected to the conserved field of prior reasoning. Genuine autoinjection requires that the system's own reasoning history be available to it as a navigable information field, and that self-evaluation traverse that field rather than produce a fresh response to a fresh prompt.

3. Architectural Requirements Derived from the Three Laws

Each law generates a specific architectural requirement. Together, the three requirements define the minimum structure of an information-native AI system.

3.1 From Conservation: Persistent Reasoning Graph

The Law of Information Conservation requires that no reasoning step, output, hypothesis, contradiction, or failed path be permanently discarded. This requirement cannot be satisfied by a context window of any size: a context window is a bounded active region, and any information that falls outside it is effectively severed from the reachable network, regardless of whether it is technically stored somewhere. The requirement is not for storage but for retrievable connectivity.

The architectural structure that satisfies this requirement is a persistent reasoning graph: a graph in which each node represents an information point generated during reasoning (a step, an output, a hypothesis, a failure, a contradiction) and each edge represents a connection between information points, including the connections that were attempted and failed. The graph grows monotonically: nodes and edges are added, never removed. The active context window is not the graph but the current traversal position within it — analogous to the relationship between a cursor position and the document it moves through.

This structure is distinct from a simple log or transcript. A log records what was generated in sequence. A reasoning graph records what was generated and how it was connected, including failed connections, contradictions between nodes, and the structural reasons that particular paths were abandoned. These relational properties are precisely what the Law of Conservation requires to be preserved: not just the content of discarded paths but their position in the connection topology.

3.2 From Connection: Topology-Aware Navigation

The Law of Connection requires that the system be capable of establishing connections between its current reasoning state and previously conserved information points. Similarity-based retrieval — the dominant paradigm in retrieval-augmented generation — is insufficient for this requirement. Similarity search returns content that resembles the query. Topology-aware navigation returns content that is structurally adjacent to the current position in the reasoning graph, regardless of surface similarity.

The distinction matters in precisely the cases where information conservation is most valuable: non-linear reasoning problems where the solution requires passing through or returning to what appeared to be a dead end. A failed path may be semantically distant from the current query but structurally adjacent — it shares a node in the reasoning graph with the current position. Similarity search will not find it. Topology-aware navigation will.

The navigation mechanism required by the Law of Connection must therefore operate on the graph structure of the conserved field, not on the semantic content of individual nodes. It must be capable of answering questions such as: what previously failed paths are adjacent to the current node? What contradictions were recorded between the current hypothesis and earlier nodes? What connections were attempted from a structurally similar position in a prior reasoning session? These are graph traversal questions, not retrieval questions.

3.3 From Autoinjection: Structural Self-Evaluation

The Law of Autoinjection requires that the system be capable of receiving itself as input. In architectural terms, this means that the system's own reasoning graph must be available to it as a navigable field that it can evaluate, connect to, and reason about. Self-evaluation is not a separate process that queries the system from outside; it is a traversal of the same conserved field that the system uses for all other reasoning.

Current approaches to AI self-evaluation — chain-of-thought verification, self-consistency sampling, meta-prompting — simulate this property without instantiating it. They produce fresh outputs in response to meta-level prompts and compare them to prior outputs. This is evaluation by a system that observes its outputs from outside, not evaluation by a system that is structurally part of the field it evaluates. The difference is precisely the difference between an external observer and an information point within the autoinjective field.

Structural self-evaluation requires that the system's evaluative operations be recorded as nodes in the same persistent reasoning graph as the outputs they evaluate, connected to those outputs by typed edges (evaluated, contradicted, confirmed, extended). The evaluation history is then itself a conserved field that subsequent reasoning can navigate. The system can observe not just what it concluded but how it evaluated prior conclusions, and whether those evaluations have been confirmed or disconfirmed by subsequent reasoning.

4. The Current State of AI Development Against These Requirements

We assess each requirement against the current state of AI research and development.

4.1 Against the Requirement for a Persistent Reasoning Graph

Current AI systems manage information through context windows, external databases, and various forms of retrieval-augmented generation. None of these satisfies the requirement for a persistent reasoning graph.

Context windows are bounded active regions. Information that falls outside the context window is severed from the reachable network. Token management — the practice of summarizing, compressing, or truncating content to fit within the window — is itself a violation of the Law of Conservation: summarization replaces an information point with a compressed approximation, and the original point, along with its relational structure, is lost. The act of token management is architecturally equivalent to annihilation, regardless of whether the summarized content is stored somewhere in a less connected form.

Vector databases store embeddings of content chunks and retrieve them by similarity. They do not store the relational structure between chunks — the connection topology of the reasoning that generated them. A failed reasoning path stored as a vector embedding has lost the information that made it a reasoning path: why it was pursued, why it failed, what it was adjacent to, and what contradiction caused its abandonment. The content is retained but the structure is discarded.

GraphRAG and related systems represent a genuine advance: they extract entities and relations from content and store them as graph structures, enabling traversal-based retrieval.

However, they operate on extracted and summarized content rather than on raw reasoning steps, and they are designed for document retrieval rather than for the conservation of live reasoning topology. They approximate the requirement without satisfying it.

Current state assessment: the requirement for a persistent reasoning graph is approximately 25–30% satisfied by the most advanced current approaches. The storage component is partially addressed; the relational structure component is largely absent; the monotonic growth requirement (nothing discarded) is not satisfied by any current system.

4.2 Against the Requirement for Topology-Aware Navigation

Navigation in current AI systems is dominated by semantic similarity search. Even systems that use graph structures for storage typically retrieve nodes by embedding similarity rather than by graph traversal from the current reasoning position. The question “what is structurally adjacent to where I am in the reasoning graph” is not answerable by any widely deployed system.

Research in neurosymbolic AI attempts to combine neural pattern recognition with symbolic relational reasoning, and some approaches enable forms of structured traversal. However, these systems operate on predefined symbolic structures rather than on dynamically generated reasoning graphs, and they do not satisfy the requirement for navigation that is responsive to the current position in a live reasoning session.

Episodic memory research, inspired by human hippocampal function, attempts to store specific experiences rather than generalized patterns and to retrieve them by structural similarity to current context. This is closer in spirit to the requirement but has not been implemented at the scale or with the relational precision that the Law of Connection demands.

Current state assessment: the requirement for topology-aware navigation is approximately 10–15% satisfied. The conceptual need is recognized in several research directions; practical implementation at the required level does not exist.

4.3 Against the Requirement for Structural Self-Evaluation

Self-evaluation in current AI systems is implemented through prompting: the system is asked to evaluate its prior output, and it produces a fresh response. This approach produces useful results in many cases but does not satisfy the architectural requirement. The evaluation is not connected to the conserved field of prior reasoning; it is a new inference that happens to take prior output as input. It does not traverse the reasoning graph; it generates a new node without structural connection to the nodes it evaluates.

Recursive self-improvement research and meta-learning approaches address related questions but focus on improving model weights through training rather than on enabling a running system to evaluate its own reasoning history as a navigable field. The distinction is between learning from past performance (weight update) and reasoning about past reasoning (structural self-evaluation).

The operator-based recursive engine described in Voronin (2025b) and implemented as a working system (Voronin, 2025d) demonstrates structural self-evaluation at the level of symbolic operator application: the system applies SToE operators to its own outputs, treats the results as new information points, and iterates. This constitutes a practical implementation of autoinjective self-evaluation within the constraints of current AI infrastructure, though it operates within session context rather than across a persistent reasoning graph.

Current state assessment: the requirement for structural self-evaluation is approximately 15–20% satisfied. Prompting-based simulation is well developed; architectural grounding in a persistent conserved field is absent.

5. The Required Vessel: A Specification

The three requirements converge on a specific architectural design. We describe this design — the vessel required for the testable prediction to be properly evaluated — in sufficient detail to constitute a research specification.

5.1 The Persistent Reasoning Graph

The core data structure is a directed multigraph $G = (V, E)$ where:

- V is the set of information point nodes. Each node represents a discrete reasoning artifact: a hypothesis, an intermediate conclusion, a failed path, a contradiction, an operator application, an evaluative judgment, or any other product of the reasoning process. Nodes are never deleted from G .
- E is the set of typed edges. Each edge connects two nodes and carries a type label from a defined vocabulary: generates (one node produces another), contradicts (two nodes are in logical tension), extends (one node elaborates another), evaluates (one node assesses another), failed-from (a node was generated from another and subsequently abandoned), and adjacent-to (two nodes share structural features without a direct generative relation).
- G grows monotonically. Every reasoning step adds nodes and edges; no step removes them. The active context window is a subgraph of G — the currently activated region — not G itself.

This structure satisfies the Law of Information Conservation: no node is annihilated; it may become unreachable from the current traversal position, but it remains in G and can be reached by appropriate navigation.

5.2 The Navigation Mechanism

Navigation through G is performed by a traversal agent that operates on graph structure rather than on content similarity. The traversal agent maintains a current position (a node or subgraph of G) and responds to navigation queries of the following types:

- Adjacency queries: given the current position, what nodes are connected to it by edges of specified types? This enables the system to find structurally adjacent nodes — including failed paths and contradictions — without requiring semantic similarity.

- Path queries: given two nodes, what paths exist between them in G , and what edge types do those paths traverse? This enables the system to understand how its current reasoning relates to prior reasoning sessions.
- Dead-end queries: given the current position, what nodes are reachable only through paths that have been marked as failed or contradicted? This enables the system to navigate deliberately to conserved but currently disconnected regions.

The traversal agent is distinct from the language model that generates reasoning content. The language model generates nodes; the traversal agent maintains G and responds to navigation queries. This separation ensures that navigation operates on graph structure rather than being approximated by in-context attention.

5.3 The Structural Self-Evaluation Loop

Self-evaluation is implemented as a periodic traversal of G in which the system takes G (or a relevant subgraph) as input, applies evaluative operators to its nodes and edges, and adds the results as new nodes connected to the evaluated nodes by evaluate edges. The self-evaluation loop has three phases:

- Traversal: the system navigates G from its current position, activating a subgraph that includes both recent reasoning and structurally adjacent prior nodes, including failed paths and contradictions.
- Evaluation: the system applies evaluative operators — drawn from the SToE operator grammar (Voronin, 2025b) or a compatible formalism — to the activated subgraph. Evaluation produces new nodes: confirmations, revisions, contradiction resolutions, or newly identified connections.
- Integration: the new nodes produced by evaluation are added to G with typed edges connecting them to the nodes they evaluate. The evaluation history is thus itself a conserved field, navigable by subsequent traversals.

This loop satisfies the Law of Autoinjection: the system receives G , which includes itself (its own prior evaluations), as input. The evaluation is not external to the field; it is a traversal that adds to the field.

6. The Testable Prediction, Restated

The testable prediction stated in Voronin (2025c) is reproduced here and restated in terms of the architectural specification:

An AI reasoning system that conserves and indexes all intermediate outputs — including failed reasoning paths, contradictions, and discarded hypotheses — will reach correct solutions to complex, multi-step problems at a higher rate than an identical system that discards such outputs, when tested on problems that require non-linear navigation through a large solution space.

In terms of the architectural specification: a system implementing the persistent reasoning graph (Section 5.1) and the topology-aware navigation mechanism (Section 5.2) will reach

correct solutions to problems requiring non-linear reasoning at a higher rate than an identical language model operating with standard context management (summarization, truncation, or simple retrieval augmentation).

The prediction is falsifiable on two levels. At the architectural level: if a system implementing the full specification described in Section 5 shows no advantage over a non-conserving system on the specified problem class, the Law of Information Conservation as applied to AI architecture requires revision. At the current-infrastructure level: even partial implementations — systems that maintain fuller reasoning traces than standard approaches without implementing the full graph structure — should show graded improvement proportional to the degree to which they satisfy the conservation requirement. This graded version of the prediction is testable with current infrastructure.

It is important to note that current AI systems cannot serve as the primary test substrate for the full prediction. The reason is structural: token management, which is necessary for any context-window-based system operating on extended reasoning tasks, constitutes a violation of the conservation requirement. Testing the prediction on a system that must truncate or summarize its reasoning history is not testing whether conservation helps; it is testing whether partial conservation helps relative to less conservation. The full test requires the vessel described in Section 5.

7. Relation to Existing Research Directions

The architectural specification derived here is related to several existing research directions, which it extends, reframes, or distinguishes itself from.

Retrieval-augmented generation (RAG) addresses the conservation requirement partially by maintaining an external store of information that the model can retrieve. The limitation, as noted in Section 4, is that RAG stores content without relational structure, and retrieval is similarity-based rather than topology-aware. The persistent reasoning graph extends RAG by requiring that relational structure be preserved and that navigation be graph-structural rather than similarity-based.

GraphRAG (Edge et al., 2024) represents the most direct current approximation of the required architecture. It constructs knowledge graphs from source documents and retrieves by graph traversal. The gap is that GraphRAG operates on pre-processed document content rather than on live reasoning topology, and it does not satisfy the monotonic growth requirement. The specification in Section 5 can be understood as a generalization of GraphRAG to the domain of live reasoning, with the additional requirements derived from the Law of Connection and the Law of Autoinjection.

Neurosymbolic AI research (Garcez et al., 2019; Marcus, 2020) pursues the combination of neural pattern recognition with symbolic relational reasoning. The specification in Section 5 is compatible with neurosymbolic approaches and can be understood as providing a foundational motivation for their development: the Laws of SToE explain why symbolic relational structure is not merely useful but necessary for a genuinely information-native AI system.

Non-well-founded set theory (Aczel, 1988), which permits circular set membership and provides the mathematical precedent for autoinjection (Voronin, 2025a), suggests a connection between the architectural requirement for structural self-evaluation and the mathematical study of self-referential structures. Systems implementing the autoinjective self-evaluation loop described in Section 5.3 instantiate, at the computational level, the self-containing structures that non-well-founded set theory formalizes mathematically.

Wheeler’s (1990) “it from bit” proposal and Floridi’s (2011) philosophy of information share with SToE the foundational commitment that information is the primitive ontological category. The specific contribution of the SToE laws is to move from this ontological claim to architectural requirements: not merely that information is fundamental, but that conservation, observer-dependent connection, and autoinjective self-containment are the structural properties that a system working with information must respect.

8. Conclusion

The characteristic failure modes of current AI systems — context loss, inability to navigate failed reasoning paths, simulated rather than structural self-evaluation — are not incidental engineering limitations. They are the predictable consequences of building AI systems without reference to the foundational laws governing information. The Law of Information Conservation, the Law of Connection, and the Law of Autoinjection, developed across the SToE research program (Voronin, 2021, 2025a, 2025b, 2025c), generate specific architectural requirements that current systems do not satisfy.

This paper has made the derivation from laws to requirements explicit, assessed the current state of AI development against each requirement, specified the vessel required to satisfy all three, and restated the testable prediction in architectural terms. The three laws function not as philosophical commentary on AI but as a formal design specification. The next generation of AI architecture — one that conserves all information points in a persistent reasoning graph, navigates that graph by topology rather than similarity, and evaluates itself structurally rather than through prompting — is derivable from these laws.

The full axiomatization of the SToE foundational laws, the formal consistency proofs for autoinjective systems, and the experimental evaluation of the testable prediction remain as future work. The working implementation described in Voronin (2025d) provides an existence proof that operator-based recursive reasoning within current AI infrastructure is feasible. The vessel described in Section 5 specifies what must be built for the full prediction to be properly tested.

The gap between current AI and information-native AI is not a gap of scale or training technique. It is a gap of foundational architecture. The three laws identify it precisely.

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